

Supplementary Material

WP-M2 Net: Wavelet-Perception Macro-Micro Dynamic 1D-CNN for Efficient Sequential Motion Recognition

Jun Cheng, Bo Chen*, Senior Member, IEEE, Zheming Wang, Member, IEEE,
Xiangming Ye, Ruidong Cheng, and Tian Wang

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Part I. Experimental Configuration and Evaluation Protocols

Table S1: Global hyperparameters, dataset-specific signal processing, and evaluation protocols.

Block	Setting	RAS-KP	ADSE	SurfMyoAiR
Panel A. Dataset, input, and signal preprocessing				
Dataset	Access status	Private cohort	Public	Public
Participants	Number of subjects	10	10	50
Recognition task	Vocabulary	12 uppercase letters	10 digits	26 uppercase letters
Trials	Balanced sample count	12,000; 1,000/class	10,000	13,000; 10/class/subject
Input	Channels, C	10	8	5
Input	Sampling rate, f_s	1000 Hz	800 Hz	2000 Hz
Input	Matrix, $C \times T$	10×2000	8×4000	5×8000
Input	Standardized duration	2 s	5 s	4 s
Filtering	Band-pass	20–450 Hz	20–380 Hz	20–850 Hz
Filtering	Band-pass design	Fourth-order Butterworth filter, applied offline.		
Filtering	Power-line suppression	50 Hz notch filter ($Q = 30$).		
Normalization	Application scope	Independent per-trial, per-channel Z-score using each channel’s within-trial mean and standard deviation.		
W-MP	Stage decomposition levels	{5, 4, 3, 2}	{6, 5, 4, 3}	{7, 6, 5, 4}
Panel B. Global WP-M2 Net architecture				
Representation	Tensor layout	$[B, C, T]$ at model input; logits $[B, K]$ at output.		
Embedding stem	Convolution sequence	Three Conv1D–BN projections: kernels $\{7, 3, 3\}$, strides $\{2, 2, 2\}$, channels $C \rightarrow 12 \rightarrow 24 \rightarrow 48$; Mish after the first two.		
Pyramid	Stage specification	Channels $\{48, 96, 144, 192\}$; depths $\{2, 2, 2, 2\}$; inter-stage depthwise Conv1D ($k = 3, s = 2$) followed by pointwise channel projection.		
W-MP/MA	Wavelet and kernels	Frozen Daubechies-4 (db4) analysis/reconstruction filters; $K_W = 9$; $K_S = 5$ with 8 outer groups.		
SE/FFN	Channel/feature mixing	SE reduction ratio = 4; FFN hidden width = $2C_i$ at stage i .		
Regularization	Stage dropout	{0.10, 0.15, 0.20, 0.25} for stages 1–4.		
Classifier	Output head	Adaptive average pooling, BN, dropout = 0.10, and a dataset-specific linear classifier.		
Panel C. Optimization, regularization, and checkpoint selection				
Optimizer	Adam	Initial learning rate = 10^{-3} ; weight decay = 10^{-3} ; $(\beta_1, \beta_2) = (0.9, 0.999)$.		
Training budget	Batch/epochs	Batch size = 8; maximum = 200 epochs.		
LR schedule	ReduceLROnPlateau	Monitor validation loss; mode = min; factor = 0.5; patience = 3 epochs.		
Early stopping	Stopping rule	Monitor validation loss; mode = min; patience = 10 epochs.		
Objective	Classification loss	Cross-entropy with label smoothing $\epsilon = 0.10$.		
Augmentation	Amplitude/Mixup	Amplitude scale $s \sim \mathcal{U}(0.8, 1.2)$ with probability 0.5; mini-batch Mixup $\alpha = 0.2$, applied afterward.		
Model selection	Checkpoint rule	Retain the checkpoint with minimum isolated validation loss; test data are excluded from all model-selection decisions.		
Panel D. Data partitioning and evaluation				
Intra-subject	Outer evaluation	Trial-wise stratified 10-fold CV; split seed = 0.		10-fold CV
Inter-subject	Outer evaluation	LOSO-CV; one participant tested per fold	LOSO-CV; one participant tested per fold	Five-fold inter-subject CV following the source study; 40/10 participants per outer split
Partitioning	Train/validation/test	Approximately 8:1:1, isolated within each evaluation split.		Five-fold outer evaluation; model selection uses an isolated validation subset of each training fold.
Efficiency	CPU latency	Intel Core i7-14700HX; batch = 1; pre-segmented 8×4000 input; 100 warm-up and 1,000 timed runs.		

Abbreviations: BN, batch normalization; CV, cross-validation; FFN, feed-forward network; LOSO, leave-one-subject-out; MA, micro aggregation; SE, squeeze-and-excitation; W-MP, wavelet macro perception.

Part II. Statistical Analysis

Table S2: Statistical scope, descriptive summaries, and complete participant-level comparisons.

Panel A. Evaluation scope and WP-M2 Net descriptive summaries					
Dataset	Protocol	Analysis unit	n	Accuracy (%)	Statistical role
RAS-KP	Inter-subject LOSO-CV	Held-out participant	10	93.18 (3.98)	Formal inference
ADSE	Inter-subject LOSO-CV	Held-out participant	10	90.42 (5.76)	Formal inference
RAS-KP	Intra-subject 10-fold CV	Fold summary	10	98.43 (0.71)	Descriptive only
ADSE	Intra-subject 10-fold CV	Fold summary	10	94.43 (1.30)	Descriptive only
SurfMyoAiR	Intra-subject 10-fold CV	Fold summary	10	88.69 (1.24)	Descriptive only
SurfMyoAiR	Inter-subject 5-fold CV	Test-fold aggregate	5	72.55 (5.11)	Descriptive only
Panel B. Complete participant-level comparisons					
Comparator	Test	Estimate Δ (pp)	95% CI	Effect	p_{Holm}
RAS-KP LOSO-CV comparisons ($n = 10$ participants)					
ResNet-18	Wilcoxon	7.81	[6.76, 12.98]	$r = 1.00$	0.007812
InceptionNet	Paired t	12.85	[9.91, 15.78]	$d_z = 3.13$	0.000035
CNN-Transformer	Paired t	14.98	[10.34, 19.61]	$d_z = 2.31$	0.000360
MSGAT-LSTM	Wilcoxon	6.15	[4.70, 8.70]	$r = 1.00$	0.007812
STGNN-LMR	Paired t	9.84	[5.73, 13.95]	$d_z = 1.71$	0.002127
STFEN (Mod.)	Paired t	2.29	[1.08, 3.52]	$d_z = 1.34$	0.007812
TCN	Paired t	10.83	[7.11, 14.55]	$d_z = 2.08$	0.000601
Transformer	Paired t	8.98	[3.66, 14.31]	$d_z = 1.21$	0.007812
BiLSTM	Paired t	13.00	[8.73, 17.28]	$d_z = 2.18$	0.000503
ADSE LOSO-CV comparisons ($n = 10$ participants)					
ResNet-18	Wilcoxon	18.91	[15.36, 29.29]	$r = 1.00$	0.007812
InceptionNet	Paired t	21.96	[16.07, 27.84]	$d_z = 2.67$	0.000115
CNN-Transformer	Paired t	21.12	[15.58, 26.67]	$d_z = 2.72$	0.000110
MSGAT-LSTM	Paired t	9.78	[7.11, 12.46]	$d_z = 2.62$	0.000119
STGNN-LMR	Wilcoxon	9.20	[8.29, 16.25]	$r = 1.00$	0.007812
STFEN	Paired t	0.77	[-0.18, 1.72]	$d_z = 0.58$	0.101454
TCN	Wilcoxon	10.68	[7.03, 20.27]	$r = 1.00$	0.007812
Transformer	Paired t	11.34	[7.80, 14.87]	$d_z = 2.30$	0.000238
BiLSTM	Paired t	18.94	[13.26, 24.63]	$d_z = 2.38$	0.000214

Notes: Panel A reports accuracy as Mean (Std), in percent. Formal inference was limited to participant-level LOSO-CV comparisons for RAS-KP and ADSE ($n = 10$); fold-level and SurfMyoAiR summaries are descriptive. In Panel B, paired differences for which the Shapiro–Wilk test did not reject normality ($\alpha = 0.05$) were analyzed using two-sided paired t -tests; otherwise, exact two-sided Wilcoxon signed-rank tests were used. Paired t -test rows report the mean difference, 95% t confidence interval, and Cohen’s d_z ; Wilcoxon rows report the Hodges–Lehmann pseudomedian, exact 95% confidence interval, and matched-pairs rank-biserial r . All estimates are WP-M2 Net minus the comparator in percentage points. Raw p -values were Holm–Bonferroni adjusted within each dataset ($m = 9$, family-wise $\alpha = 0.05$); bold values indicate $p_{\text{Holm}} < 0.05$.

Part III. Robustness to Simulated Signal Degradation

Experimental protocol. Let $X_i \in \mathbb{R}^{C \times T}$ and y_i denote the i th preprocessed test trial and its label, respectively. Each degradation operator D_ξ , parameterized by condition and severity ξ , was applied immediately before inference. The trained model f_θ remained fixed, and the sampling rate, channel order, sequence length, and tensor dimensions were unchanged. Accuracy was computed over the complete perturbed test set:

$$\text{Acc}(\xi) = \frac{100}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_k f_\theta(D_\xi(X_i))_k = y_i \right]. \quad (1)$$

Random signal masking (channel and temporal) was evaluated on RAS-KP and ADSE, whereas the structured conditions were evaluated on RAS-KP. No model was adapted; every condition was evaluated without retraining.

Random Signal Masking. For random channel masking, a set $\Omega_m \subseteq \{1, \dots, C\}$ of exactly m channels was sampled without replacement for each test trial. Complete traces of the selected channels were set to the normalized zero level:

$$\widehat{X}_{c,t}^{(\text{ch})} = \begin{cases} 0, & c \in \Omega_m, \\ X_{c,t}, & c \notin \Omega_m, \end{cases} \quad |\Omega_m| = m. \quad (2)$$

For random temporal masking, a set $\Psi_q \subseteq \{1, \dots, T\}$ was sampled without replacement, and the corresponding time indices were set to zero simultaneously across all channels:

$$\widehat{X}_{c,t}^{(\text{tmp})} = \begin{cases} 0, & t \in \Psi_q, \\ X_{c,t}, & t \notin \Psi_q, \end{cases} \quad |\Psi_q| = \lfloor qT \rfloor. \quad (3)$$

Thus, m controls the number of missing electrodes and q controls the proportion of missing multichannel time samples.

Simulated Structured Degradation. The motion-artifact and muscle-fatigue simulations used $\rho \in \{0, 0.1, \dots, 1.0\}$, where $\rho = 0$ denotes the clean signal and $\rho = 1$ the maximum simulated intensity.

Simulated Motion Artifacts. A global moving root-mean-square (RMS) envelope located the peak activation in each trial. A Gaussian burst was then centered at that location:

$$e(t) = \text{RMS}_{W_e}\{X\}(t), \quad t^* = \arg \max_t e(t), \quad g_\rho(t) = \rho \exp \left[-\frac{(t - t^*)^2}{2\sigma^2} \right]. \quad (4)$$

With $A = \max_{c,t} |X_{c,t}|$, a fixed carrier $h(t)$, and fixed channel scaling a_c , the perturbed signal was

$$\widehat{X}_{c,t}^{(\text{art})} = X_{c,t} + \gamma A a_c g_\rho(t) h(t), \quad \gamma = 5.0. \quad (5)$$

Thus, ρ controlled only the artifact intensity; the burst duration, carrier, and channel scaling were held fixed.

Simulated Muscle Fatigue. Fatigue-like degradation combined severity-dependent spectral reweighting with activity-weighted progressive amplitude scaling. A centered Hann-window short-time Fourier transform (STFT) was computed for each channel and reweighted as

$$S_c(k, j) = \text{STFT}_{\text{Hann}}\{X_c\}(k, j), \quad N_{\text{FFT}} = 256, \quad H = 64, \quad (6)$$

$$\widetilde{S}_c(k, j; \rho) = W_c(k, j; \rho) S_c(k, j).$$

Let R_c be the within-trial RMS of channel c , and let $\bar{R} = C^{-1} \sum_c R_c$; both were floored at 10^{-6} . With normalized frequency $u_k = k/(N_{\text{FFT}}/2)$, the exact implementation-level weighting rule was

$$s_c(\rho) = \text{clip}_{[0, 1.35]} \left\{ \rho \left[0.55 + 0.45 \text{clip}_{[0.35, 2.2]} \left(\frac{R_c}{\bar{R}} \right) \right] \right\},$$

$$r_j = 0.35 + 0.65 \frac{j}{J-1}, \quad a_{c,j} = s_c(\rho) r_j, \quad u_k = \frac{k}{N_{\text{FFT}}/2},$$

$$W_c(k, j; \rho) = \text{clip}_{[0.04, 1.90]} \left\{ \left[1 + 0.85 a_{c,j} e^{-((u_k - 0.08)/0.18)^2} \right] \left[1 - 0.05 a_{c,j} e^{-((u_k - 0.22)/0.12)^2} \right] \right. \\ \left. \times \left[1 - 1.35 a_{c,j} \text{sigm} \left(\frac{u_k - 0.24}{0.07} \right) \right] \left[1 - 0.65 a_{c,j} \text{sigm} \left(\frac{u_k - 0.55}{0.08} \right) \right] \right\}. \quad (7)$$

Here, J is the number of STFT frames and $\text{sigm}(z) = (1 + e^{-z})^{-1}$, and $\text{clip}_{[a,b]}(z) = \min\{\max(z, a), b\}$. Because W_c is a positive real-valued gain, the operation retained the original STFT phase. After the inverse short-time Fourier transform (ISTFT), deterministic smoothing, partial RMS stabilization, and weak modulation were applied, followed by progressive amplitude scaling:

$$\widehat{X}_{c,t}^{(\text{fat})} = [1 + \eta s_c(\rho) v(t)] \mathcal{P} \{ \text{ISTFT}[\widetilde{S}_c(\rho)] \}, \quad \eta = 0.70. \quad (8)$$

Here, $v(t) = 0.20 + 0.80t/(T-1)$, $t = 0, \dots, T-1$, and $\mathcal{P}\{\cdot\}$ denotes deterministic Gaussian smoothing, partial RMS stabilization, and 9-Hz modulation; selected fixed settings are summarized in Table S3. The original sequence length was preserved.

Table S3: Key parameters for the signal-degradation experiments.

Condition	Dataset and severity	Fixed operator parameters
Random channel masking	RAS-KP: $C = 10, T = 2000, m \in \{0, 1, \dots, 5\}$; ADSE: $C = 8, T = 4000, m \in \{0, 1, \dots, 4\}$	Exactly m channels sampled without replacement; selected traces set to normalized zero
Random temporal masking	RAS-KP and ADSE: $q \in \{0, 0.05, \dots, 0.50\}$	$ \Psi_q = \lfloor qT \rfloor$; indices sampled without replacement and masked across all channels
Simulated motion artifacts	RAS-KP: $C = 10, T = 2000; \rho \in \{0, 0.1, \dots, 1.0\}$	$f_s = 1000$ Hz; RMS envelope 50 ms; burst duration 300 ms ($\sigma = 50$ samples); $\gamma = 5.0$; carrier frequencies 18/43/91 Hz with weights 0.55/0.30/0.15; channel-gain modulation 0.15
Simulated muscle fatigue	RAS-KP: $C = 10, T = 2000; \rho \in \{0, 0.1, \dots, 1.0\}$	$f_s = 1000$ Hz; Hann STFT $N_{\text{FFT}} = 256$, hop size 64; normalized-frequency profiles (center/scale, coefficient): Gaussian (0.08/0.18, +0.85), Gaussian (0.22/0.12, -0.05), sigmoid (0.24/0.07, -1.35), and sigmoid (0.55/0.08, -0.65); frame ramp 0.35-1.00; weight clamp [0.04, 1.90]; 31-sample Gaussian smoothing ($\sigma = 7$); 9-Hz modulation 0.16; $\eta = 0.70$

Note: The listed values define fixed simulation settings and were held constant across models. The spectral-weighting rule and all listed operator parameters were fixed before model comparison and were not optimized using test outcomes. These settings should not be interpreted as subject-specific physiological parameters. All models were evaluated using the same operator configurations and severity levels.